Biological knowledge shortfalls of Laboulbeniomycetes as a case study for poorly studied taxa



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Introduction

- **Ecological knowledge shortfalls** inhibit a full understanding of **biodiversity patterns and processes**, and impact **conservation efforts**
- Most empirical research on biological shortfalls has focused on **vertebrate taxa**
- **Progress has been slow** with addressing knowledge shortfalls in **less conspicuous** or **poorly studied taxa**



Objective

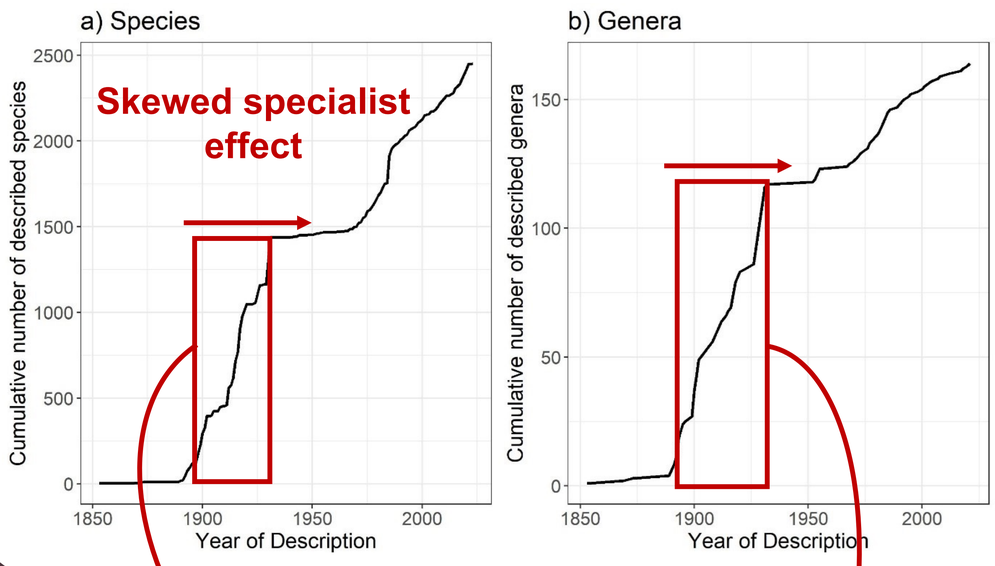
- Explore how **four shortfalls** (Linnean, Wallacen, Latimerian, and Scottian) impact the **conservation** of a model group of **poorly studied organisms**: **Laboulbeniomycetes** (Labouls)

Why Laboulbeniomycetes?

- **Fungi obligatorily associated with arthropods**, **historically neglected** due to practical constraints, recent work based on DNA indicates **high levels of cryptic diversity**.

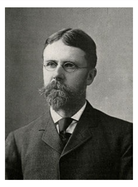
Linnean shortfall (undescribed species)

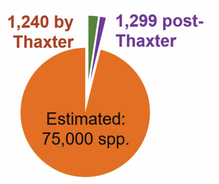
- Increased sampling effort **still returns new species**, and the "actual" number of species is likely even **higher than previously estimated**



Dr. Roland Thaxter

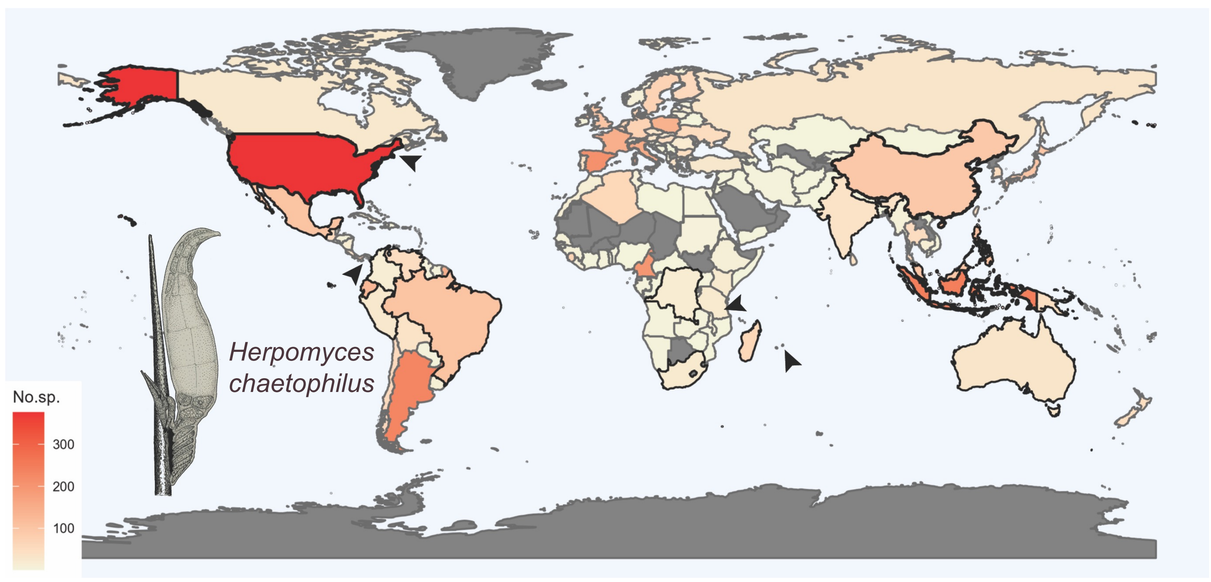
- Active 1891-1931
- 92-132 years ago





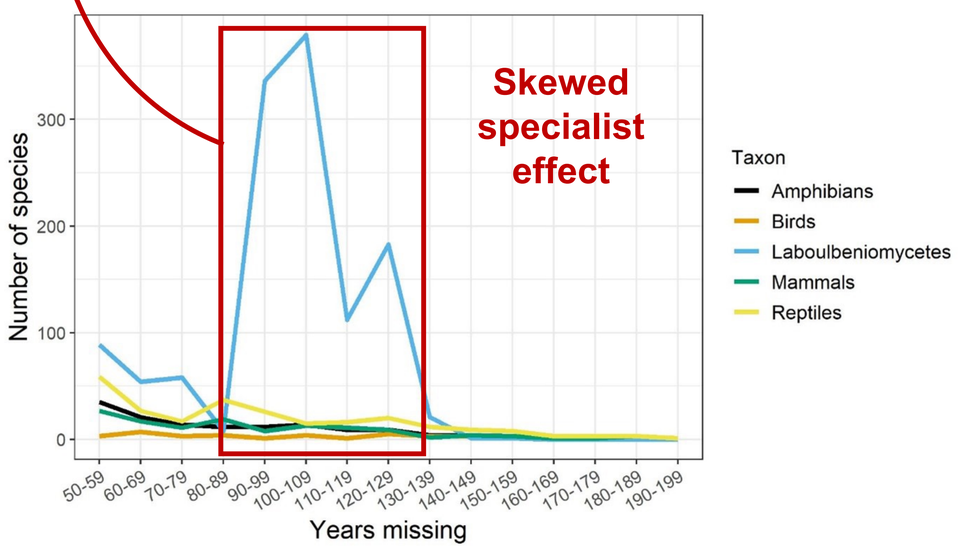
Wallacean shortfall (distribution completeness)

- We see a **distributional unevenness**, and **many taxa have disjunct distributions**



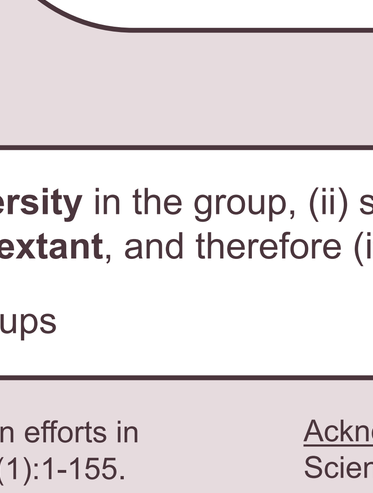
Latimerian shortfall ("last seen")

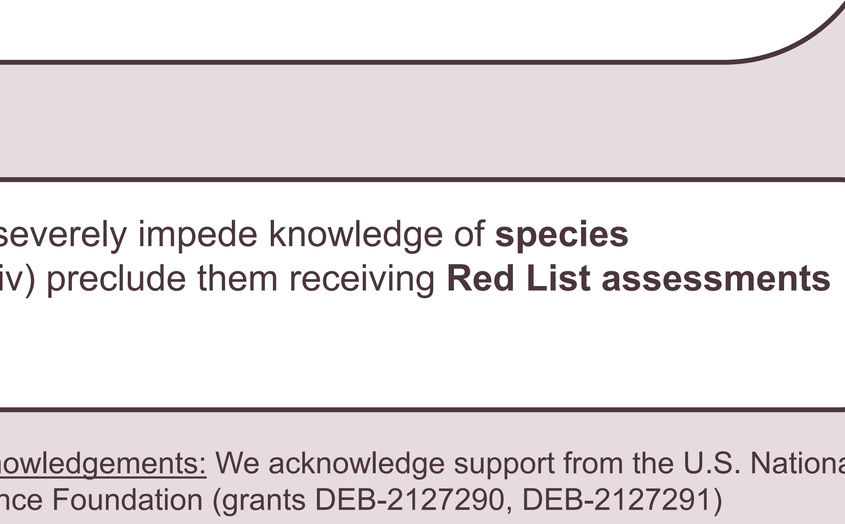
- **Half species** have no observation in **50+ years after description**



Scottian shortfall (IUCN assessments)

- **Newly-introduced** after British conservationist **Sir Peter Markham Scott** (1909-1989)
- **No Red List assessments** have been made for Labouls





Conclusion

- Biological shortfalls (i) prevent accurate **modeling of overall diversity** in the group, (ii) severely impede knowledge of **species distributions**, (iii) inhibit knowledge of **whether species remain extant**, and therefore (iv) preclude them receiving **Red List assessments**
- Reflects the **lack of specialist taxonomists** in poorly studied groups

Reference: Haelewaters et al. 2024. Biological knowledge shortfalls impede conservation efforts in poorly studied taxa – a case study of Laboulbeniomycetes. Journal of Biogeography, 51(1):1-155.

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